M. ADITYA PATRA

Pune, Maharashtra, India | Email | Portfolio | GitHub | LinkedIn

Summary

B.Tech Computer Science student who enjoys building machine learning systems end to end, from raw public data through to tested models that other people can run and use. Most of my work is in space and orbital data, motorsport analytics and environmental time series. The parts I care about most are careful validation, finding out where a model's accuracy really comes from, and turning results into working tools such as APIs and dashboards. Seeking an ML Engineer or Data Science internship.

Technical Skills

Languages	Python, SQL (MySQL, PostgreSQL), Java, C
ML & Data	scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, SHAP, pandas, NumPy, SciPy, statsmodels, Prophet, lifelines, PySpark
Methods	Feature engineering, group-aware and geographic cross-validation, leakage testing, time-series forecasting, survival analysis (Kaplan–Meier, Cox PH), hypothesis testing (Welch t-test, Chi-square, Diebold–Mariano), probability calibration, drift monitoring, clustering (K-Means, DBSCAN)
Apps & Viz	Plotly Dash, Streamlit, Next.js, CesiumJS, Plotly, Matplotlib
Engineering	FastAPI, ONNX, Docker, GitHub Actions, pytest, Git, MySQL Workbench, VS Code

Experience

Data Analyst & Machine Learning Intern June 2026 – August 2026
CloudExperts.co *Remote*

- Built a Nifty 500 stock screener and a broker automation layer on Zerodha's Kite Connect API (GTT/OCO orders, WebSocket fill detection, trailing stop-loss).
- Built an anomaly-detection pipeline for warehouse sensor data and a command-line tool for searching words across PDF documents.
- Reviewed production code for a RAG web application built on React/TypeScript and AWS Lambda.

Technical Team Member Oct 2026 – Present
Tech For Need, DY Patil International University *Pune*

- Joined the technical team at the start of October 2026.

Logistics & Organizing Member 2025 – 2026
Concordia Club, DY Patil International University *Pune*

- Coordinated logistics and participant management for 3+ university events, including Model United Nations, with 100+ participants across concurrent sessions.

Projects

OrbitGuard: Orbital Collision Screening & Risk Forecasting
Python, SGP4, scikit-learn, FastAPI, PostgreSQL, Docker, GitHub Actions, CesiumJS GitHub | Live Dashboard

- Screens ∼14,400 tracked satellites and debris objects for close approaches every 6 hours. It uses CelesTrak data, SGP4 propagation and a KD-tree search padded for closing speed, so head-on encounters can't be missed (covered by a unit test). A relative-motion pre-filter made screening ∼6× faster with identical results.
- Estimated collision probability without published covariance by measuring how element-set error grows in each population (Starlink goes stale ∼50× faster than debris). With that, events at $P_c \geq 10^{-4}$ fell from 97 to 7.
- Trained a risk forecaster on ESA's Collision Avoidance Challenge data (162,634 messages) that predicts final risk two days before closest approach. It scored F2 0.749 on the held-out test set, a statistical tie with a baseline that only 12 of 97 competition teams beat.
- Results are served through FastAPI and PostgreSQL (∼60 ms filtered queries over a 46,000-conjunction run), with a validation and drift-check pipeline, Docker, a scheduled GitHub Actions run and a live 3-D CesiumJS dashboard.

Lunar Crater Detection
Python, PyTorch, U-Net, ONNX, FastAPI, pytest, GitHub Actions GitHub

- Detected impact craters in LRO/LOLA elevation tiles labelled from the Robbins catalogue. A geographic train/test split with buffer zones meant test terrain was never seen in training or tuning.
- Compared a classical radial-slope filter, a U-Net and a multi-scale U-Net. The served model reached F1 0.539 on 429 held-out tiles (baseline 0.369) and 0.562 on craters over 8 km, after a separate large-crater output fixed false merged detections.

- Exported to ONNX and served behind FastAPI. Each crater comes back with latitude, longitude, diameter and calibrated confidence (mean error 0.014), and a `/monitoring` endpoint flags input and prediction drift. CI runs on every pull request.

Exoplanet Candidate Classification & Habitability
Python, XGBoost, TensorFlow, SHAP, Streamlit, Next.js GitHub

- Classified Kepler's 1,977 unresolved candidates with six models trained on 7,587 labelled objects, using 5-fold cross-validation grouped by host star. The best F1 was 0.9938 (Gradient Boosting), and 1,267 candidates were predicted likely real. An ablation across three feature tiers showed NASA's vetting flags add 2–5 F1 points over raw physics.
- Scored 7,004 planets for habitable-zone status (Kopparapu bounds) and a simplified Earth Similarity Index. 3.21% fall in the conservative habitable zone, and 38 are both likely rocky and inside a zone.

BOXBOX: F1 Race Centre & Strategy Lab
Python, FastF1, OpenF1, scikit-learn, lifelines, SHAP, Plotly Dash GitHub

- Race Centre: replays the latest race lap by lap from free OpenF1/Jolpica data, tracks both championships and loads each new race automatically once its data is published.
- Strategy Lab on the full 2024 season (23 circuits): tyre-degradation models for each circuit and compound, chosen by cross-validated RMSE (Random Forest won 39 of 59, Ridge 20). Kaplan–Meier and Cox models cover stint length, and every 1-, 2- and 3-stop plan is simulated and ranked into Plans A/B/C with risk scores.
- Six ML models (stint length, compound choice, pit window, undercut viability and more) are explained with SHAP. DBSCAN independently checked the manual data cleaning, and fixing two upstream bugs restored real risk differences between circuits.

India AQI Forecasting
Python, PyTorch, statsmodels, Prophet, XGBoost, SHAP, Streamlit GitHub

- Forecast next-day AQI for five Indian cities from CPCB data (2015–2020). Compared 10 models, from persistence baselines to SARIMAX, Prophet, XGBoost, LSTM and GRU, on the same rows, with a chronological split and leakage tests for every model family.
- The LSTM cut next-day error by 19% against persistence (MAE 14.7 vs 18.2, Diebold–Mariano $p < 0.001$) and roughly halved error on severe-pollution days (42.6 → 21.4). SHAP showed the gain came from the previous day's pollutant readings.

Kepler Exoplanet Classification
Python, scikit-learn, XGBoost, SHAP, Streamlit GitHub

- Classified 9,564 Kepler objects as planets or false positives (best F1 ≈ 0.996), then audited the score. On raw physical measurements alone, F1 drops to ≈ 0.94, which shows how much of the result came from NASA's vetting flags.
- Found and fixed six methodology issues, including SMOTE and imputer leakage inside cross-validation and host-star leakage in the train/test split. Built a five-tab Streamlit dashboard.

Football Archive Relational Database
MySQL GitHub

- Designed a normalised (3NF) MySQL database of 11 tables covering the Premier League, La Liga and Bundesliga, enforced with 16 foreign keys and STRICT mode.
- Built 5 analytical views, 3 stored procedures and 12 queries, including xG over/under-performance, goal contributions per 90 and a discipline index.

Other projects: Heart Disease Analysis (Random Forest, ROC-AUC 0.909, 84% accuracy) | Network Intrusion Detection (KDD Cup 99, hypothesis tests, three benchmarked classifiers)

In progress: xG-Spark (expected-goals model on PySpark with a Parquet lakehouse) | DataViz (no-code charting app in Streamlit)

Education

B.Tech in Computer Science & Engineering 2024 – 2028
DY Patil International University, Pune

Class 12 2023
Army Public School, Bhopal

Certifications

Data Mining in Python, University of Michigan (2024) | Technology Commercialization, University of Rochester (2024)

Languages

English (Fluent), Hindi (Fluent), Odia (Native), French (Basic)